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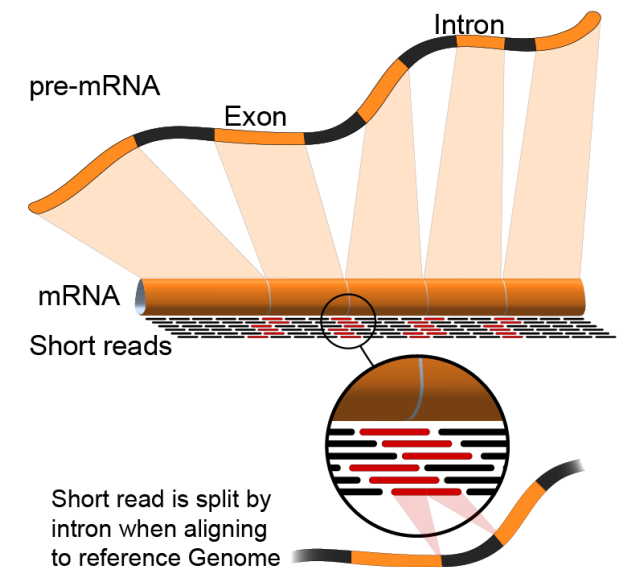
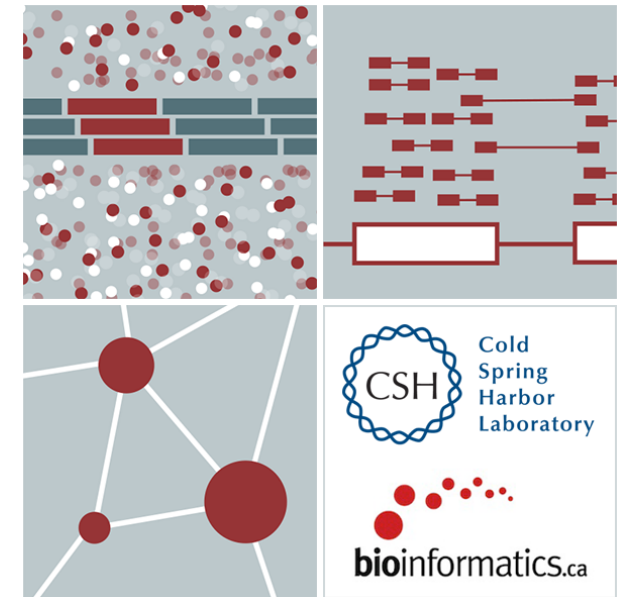
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# Module 1: Introduction to RNA sequencing

Malachi Griffith & Obi Griffith

RNA-seq Analysis  
June 22-24, 2026



# Learning objectives of the course



- **Module 1: Introduction to RNA Sequencing**
- Module 2: Alignment and Visualization
- Module 3: Expression and Differential Expression
- Module 4: Alignment Free Expression Estimation
- Tutorials
  - Provide a working example of an RNA-seq analysis pipeline
  - Run in a ‘reasonable’ amount of time with modest computer resources
  - Self contained, self explanatory, portable

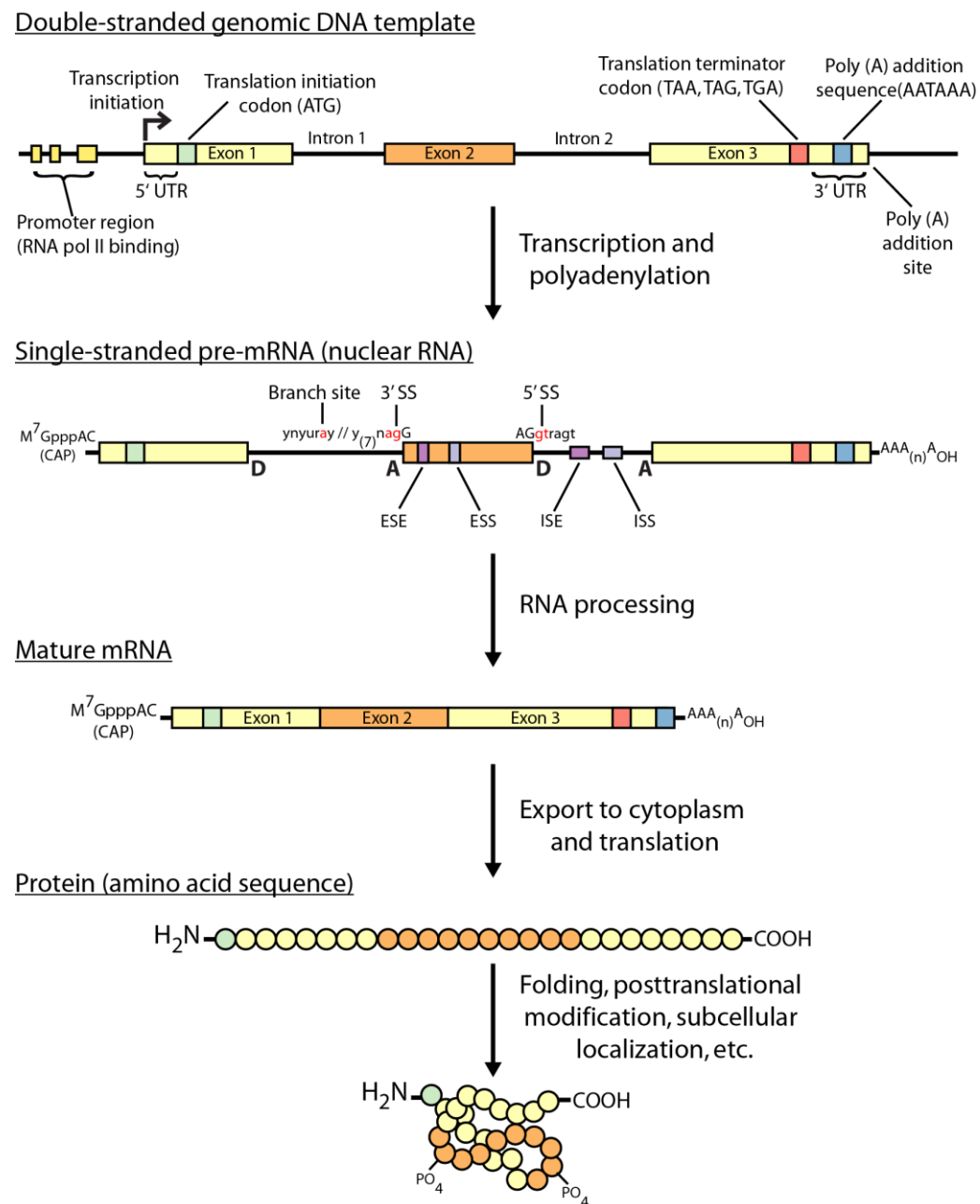


# Learning objectives of module 1

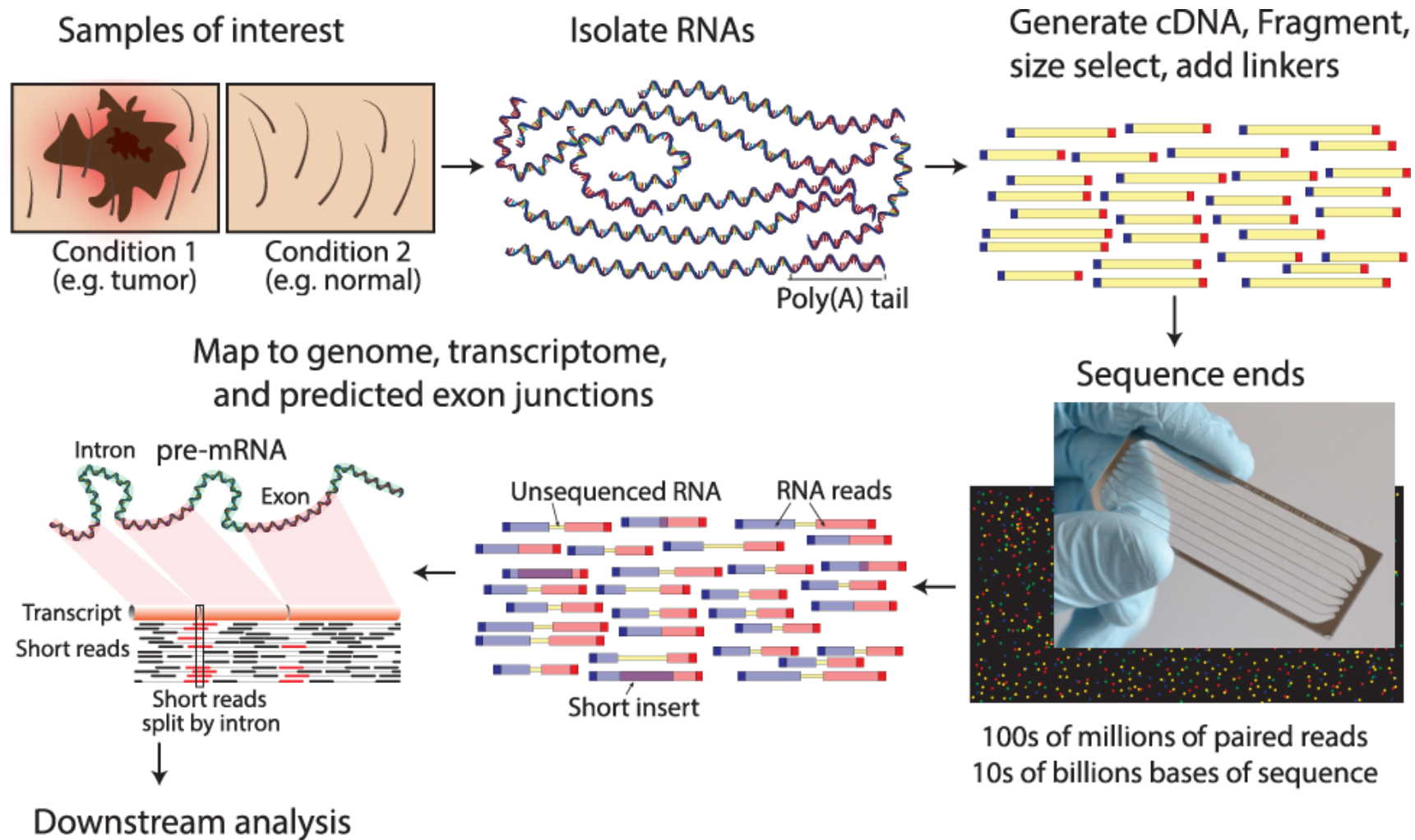
- Introduction to the theory and practice of RNA sequencing (RNA-seq) analysis
  - Background molecular biology
  - Challenges specific to RNA-seq
  - General goals and themes of RNA-seq analysis workflows
  - Common technical questions related to RNA-seq analysis
  - Introduction to the RNA-seq hands on tutorial



Does it differ depending on the sequencing platform? Or for bulk vs single cell sequencing?



# Overview of an RNA seq experiment





# MPS (NGS) Platforms: Illumina is currently dominant

## Production-scale sequencers

| Key specifications |  |  |  |                                                                             |
|--------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|                    | <a href="#">NextSeq 1000 and 2000 Systems</a>                                      | <a href="#">NovaSeq 6000 System</a>                                                 | <a href="#">NovaSeq X Series</a>                                                    |                                                                             |
|                    | Max output per flow cell                                                           | 540 Gb <sup>a</sup>                                                                 | 3 Tb <sup>b</sup>                                                                   | 8 Tb <sup>c</sup>                                                           |
|                    | Run time (range) <sup>d</sup>                                                      | ~8–44 hr                                                                            | ~13–44 hr                                                                           | ~17–48 hr                                                                   |
|                    | Max reads per run (single reads)                                                   | 1.8B <sup>a</sup>                                                                   | 10B (single flow cell) <sup>b</sup><br>20B (dual flow cells)                        | 26B (single flow cell) <sup>c</sup><br>52B (dual flow cells) <sup>c,e</sup> |
| Max read length    | 2 × 300 bp                                                                         | 2 × 250 bp                                                                          | 2 × 150 bp                                                                          |                                                                             |

- High accuracy, range of capacity and throughput (low cost per base)
- Slightly longer read lengths on some platforms

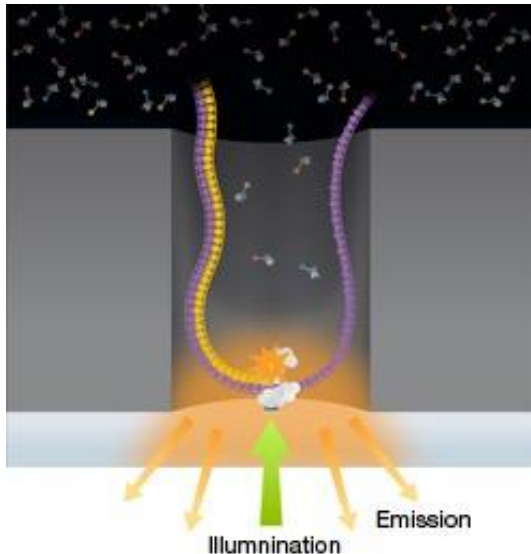


# Next-next (3rd) generation sequencing platforms

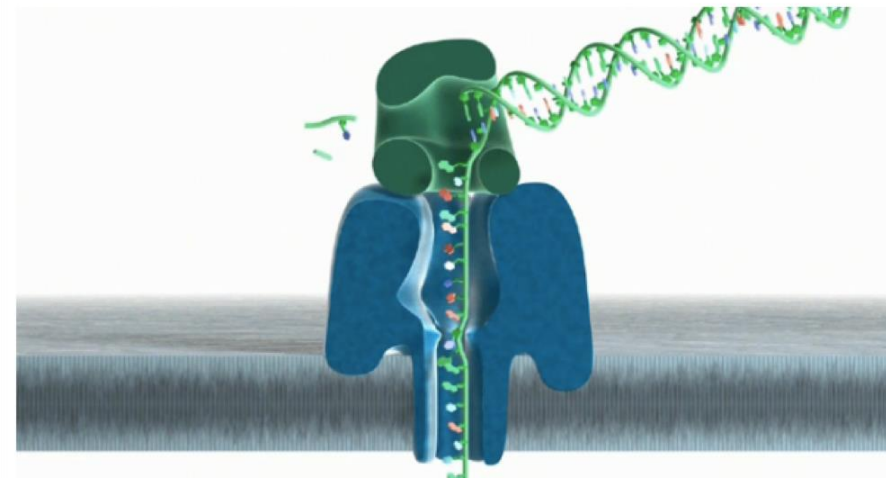


Defining Characteristics: Long reads (10-100 kb) from single molecules.

**Pacific Biosciences:** watching a polymerase synthesize DNA/cDNA in real time



**Oxford Nanopore:** Translocating DNA/RNA through a nanopore with electrode-based detection



**The promise:** Long reads will allow us to accurately sequence and assemble whole human genomes, from scratch, without using the reference genome.

**Status:** Currently limited by lower throughput, higher base error rate and higher cost. 3rd generation technologies have proven useful, but generally for niche applications so far.

# Challenges

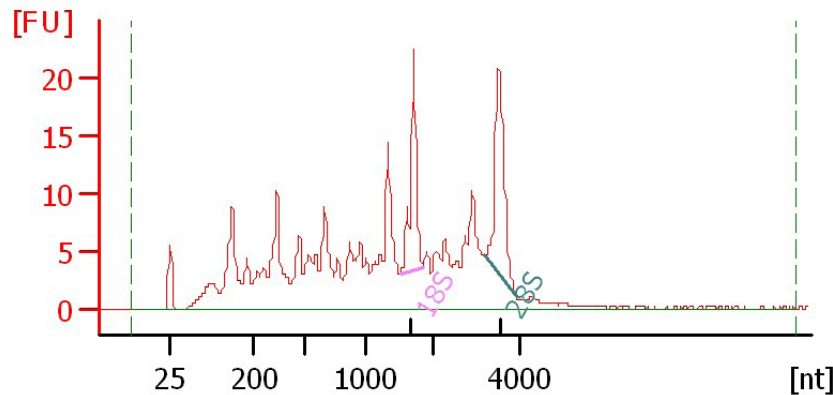


- Sample
  - Purity?, quantity?, quality?
- RNAs consist of small exons that may be separated by large introns
  - Mapping reads to genome is challenging
- The relative abundance of RNAs vary wildly
  - $10^5 - 10^7$  orders of magnitude
  - Since RNA sequencing works by random sampling, a small fraction of highly expressed genes may consume the majority of reads
  - Ribosomal and mitochondrial genes
- RNAs come in a wide range of sizes
  - Small RNAs must be captured separately
  - PolyA selection of large RNAs may result in 3' end bias
- RNA is fragile compared to DNA (easily degraded)

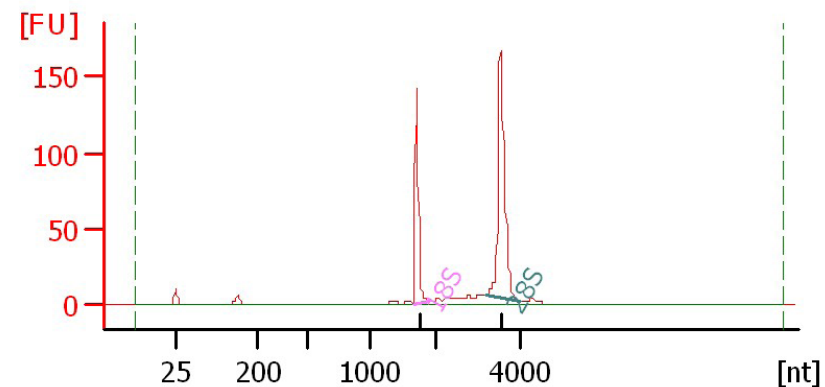
# Agilent example / interpretation



- <https://goo.gl/uC5a3C>
- 'RIN' = RNA integrity number
  - 0 (bad) to 10 (good)



RIN = 6.0



RIN = 10



# There are many RNA-seq library construction strategies

- Total RNA versus polyA+ RNA?
- Ribo-reduction?
- Size selection (before and/or after cDNA synthesis)
  - Small RNAs (microRNAs) vs. large RNAs?
  - A narrow fragment size distribution vs. a broad one?
- Linear amplification?
- Stranded vs. un-stranded libraries
- Exome captured vs. un-captured
- Library normalization?
- These details can affect analysis strategy
  - Especially comparisons between libraries

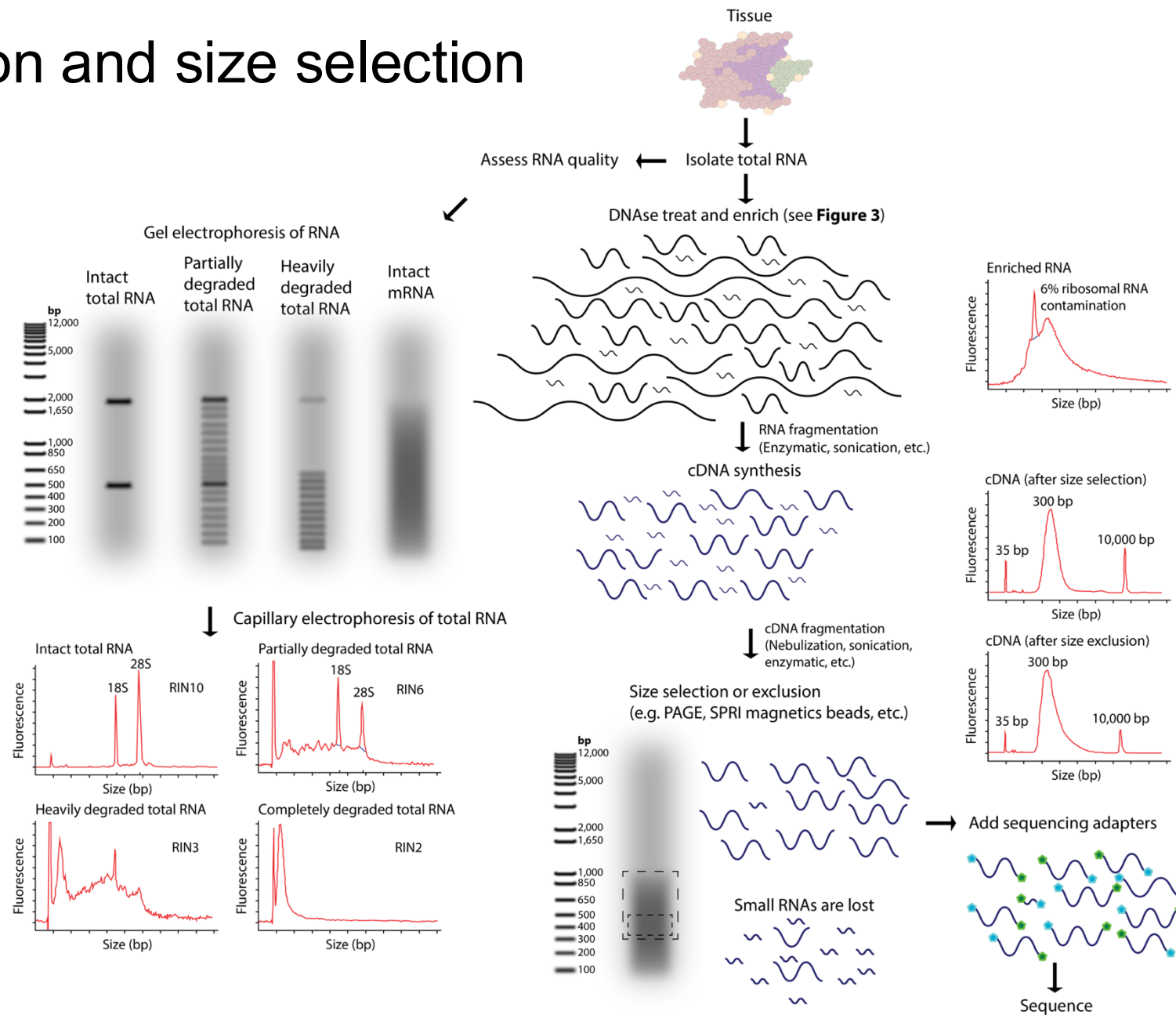


# Ordering RNA-seq data, “coverage”, and cost?

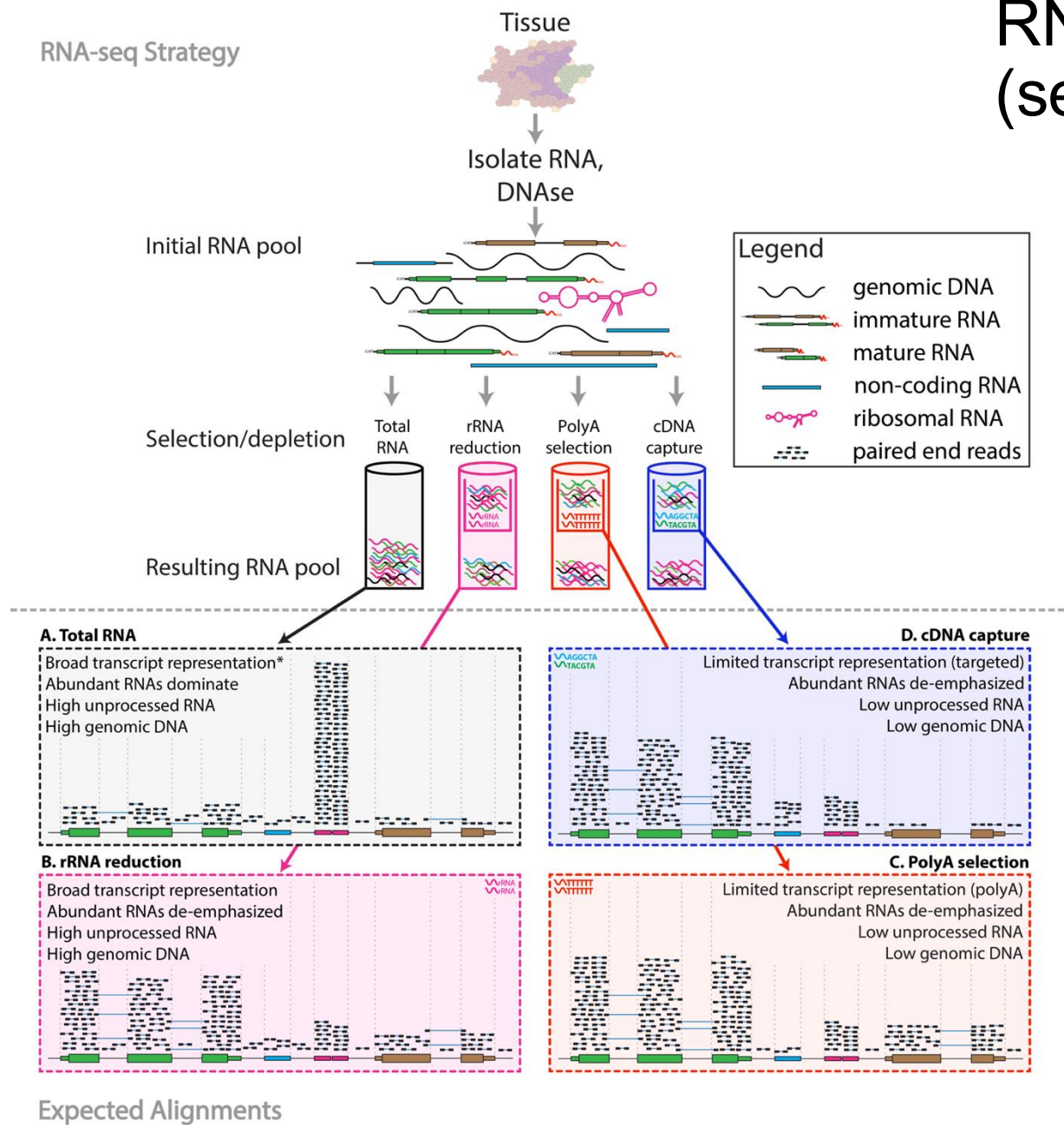
| RNA-seq full service, cost per sample <sup>a</sup> | < 12 Samples | ≥ 2500 Samples* | ~Targeted Coverage |
|----------------------------------------------------|--------------|-----------------|--------------------|
| PolyA selection                                    | \$287        | \$215           | 30M reads          |
| Ribosomal depletion, RiboErase (H/M/R)             | \$297        | \$226           | 30M reads          |
| Ribosomal depletion, FastSelect (H/M/R)            | \$268        | \$205           | 30M reads          |
| Ribosomal depletion, FastSelect (H/M/R+Globin)     | \$272        | \$213           | 30M reads          |
| Ribosomal depletion, Watchmaker (H/M/R+Globin)     | \$291        | NA              | 30M reads          |
| Low input - Takara SMARTseq mRNA                   | \$267        | \$203           | 30M reads          |
| Low input - Sigma Seqplex                          | \$273        | \$202           | 30M reads          |

- An example menu from a sequencing core facility
- Options primarily relate to method of enrichment and input amounts
- “Coverage” is a non-intuitive concept in bulk-RNAseq.
  - 30M reads is sufficient for gene abundance estimation (increase for other applications)

# Fragmentation and size selection



# RNA sequence enrichment (selection/depletion)

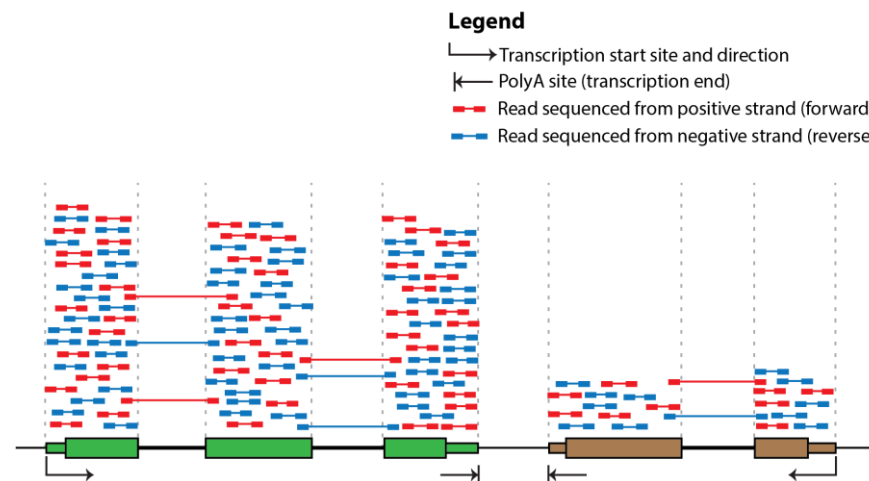




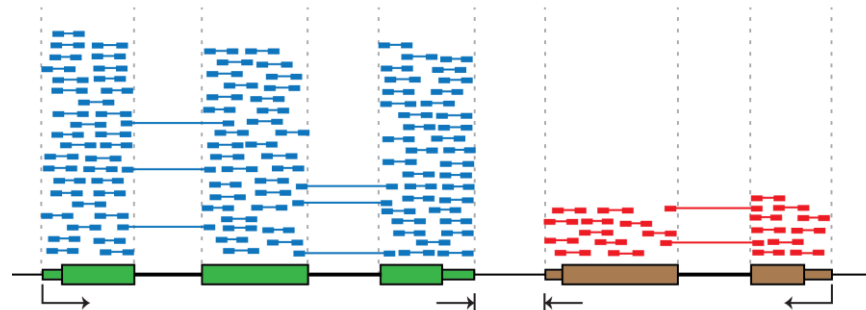


# Stranded vs. unstranded

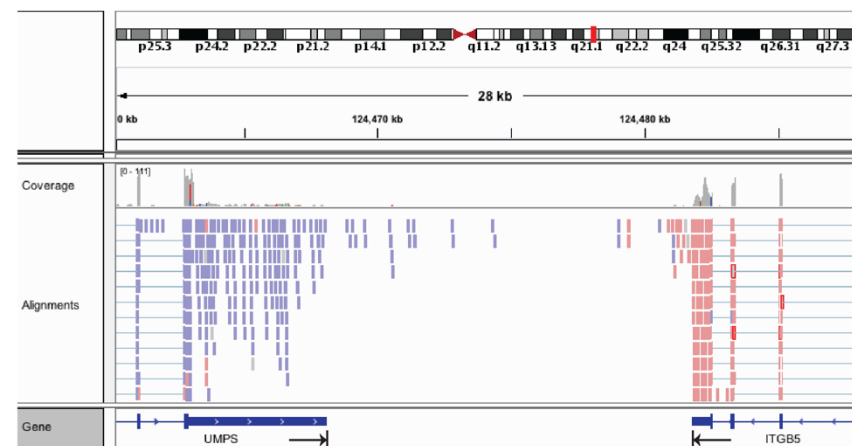
## A. Depiction of cDNA fragments from an unstranded library



## B. Depiction of cDNA fragments from a stranded library



## C. Viewing strand of aligned reads in IGV



<https://rnabio.org/module-09-appendix/0009/12/01/StrandSettings/>  
(detailed discussion and cheat sheet)



# Common analysis goals of RNA-Seq analysis (what can you ask of the data?)



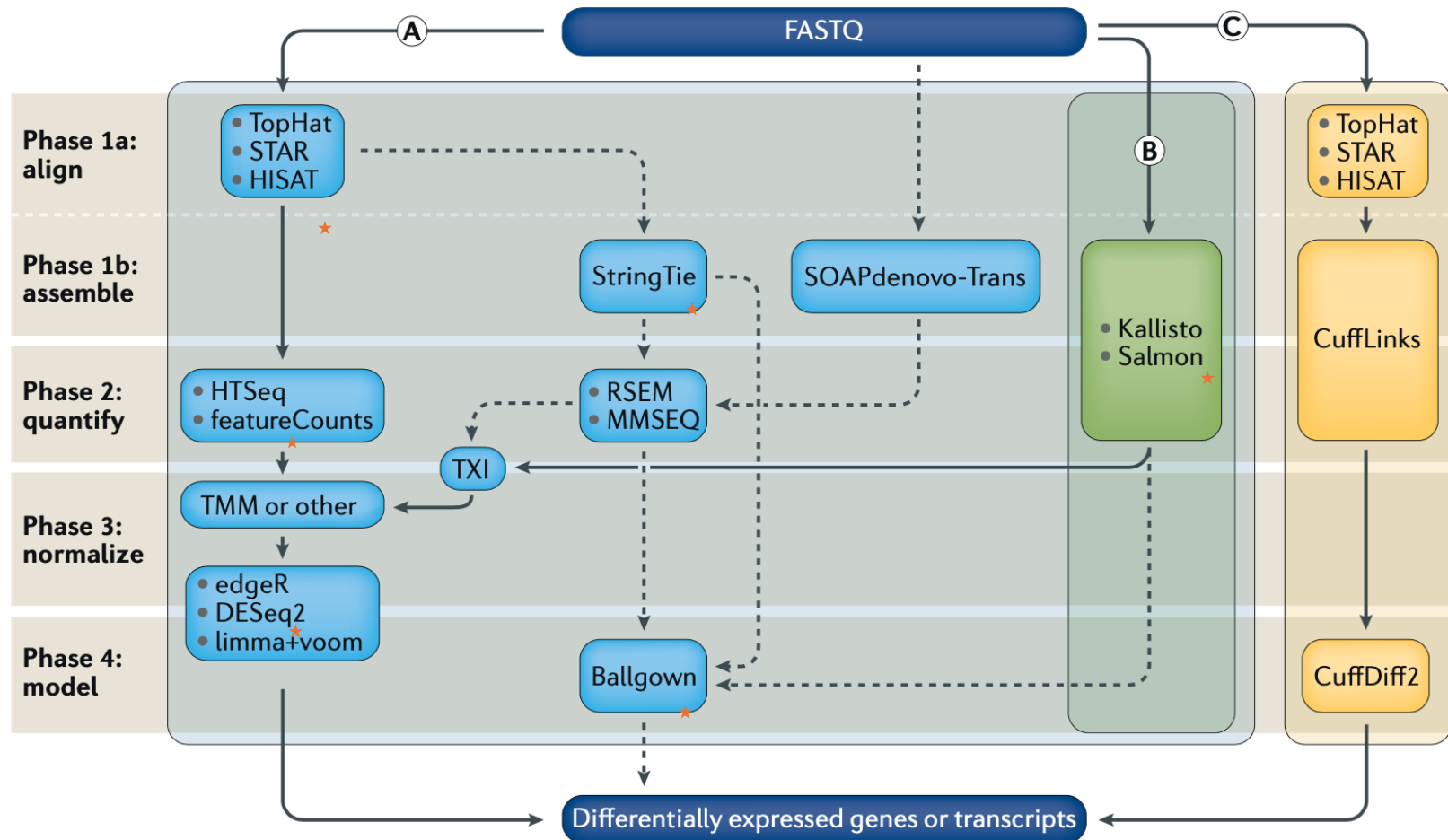
- Gene expression and differential expression
- Alternative expression analysis
- Transcript discovery and annotation
- Allele specific expression
  - Relating to SNPs or mutations
- Mutation discovery
- Fusion detection
- RNA editing

# General themes of RNA-seq workflows

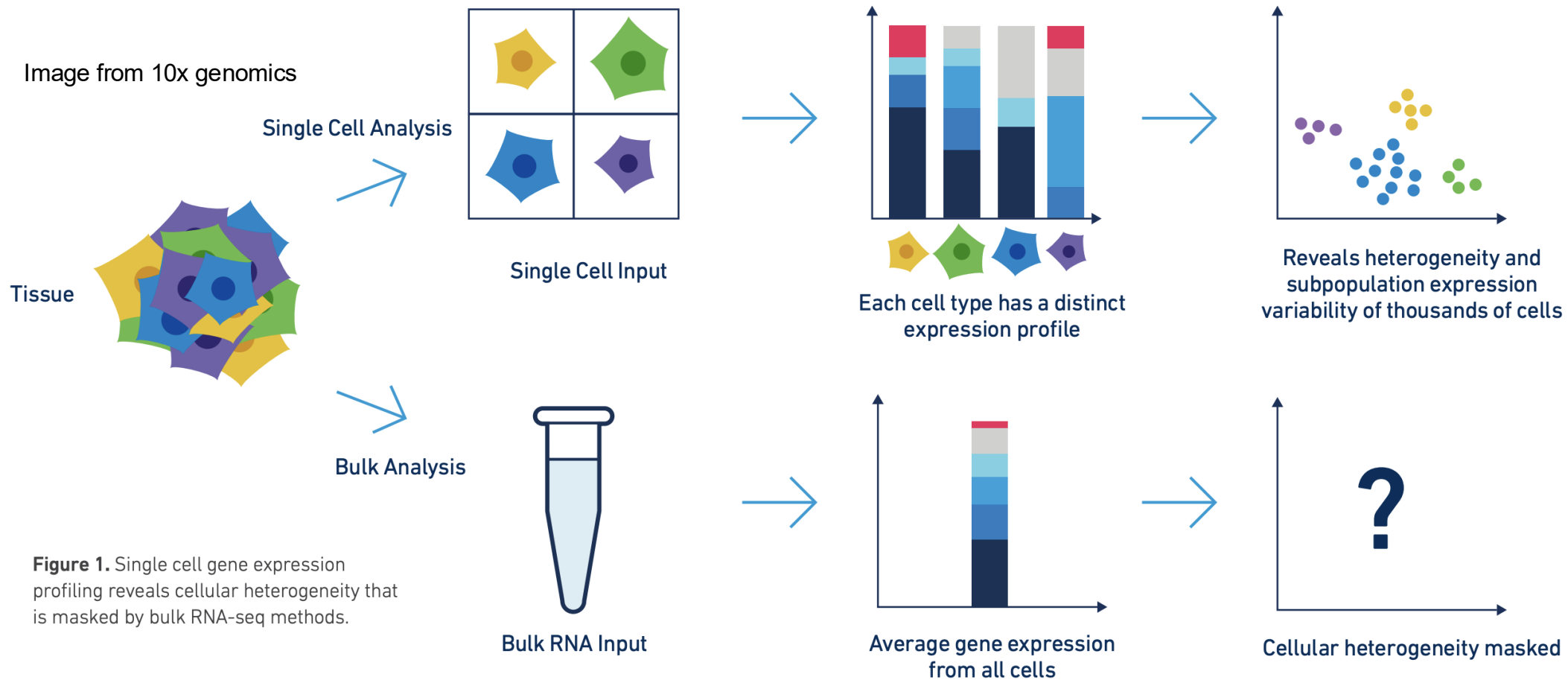


- Each type of RNA-seq analysis has distinct requirements and challenges but also a common theme:
  1. Obtain raw data (convert format)
  2. Align/assemble reads
  3. Process alignment with a tool specific to the goal
    - e.g. ‘cufflinks’ for expression analysis, ‘defuse’ for fusion detection, etc.
  4. Post process
    - Import into downstream software (R, Matlab, Cytoscape, Ingenuity, etc.)
  5. Summarize and visualize
    - Create gene lists, prioritize candidates for validation, etc.

# Examples of RNA-seq data analysis workflows for differential gene expression



# Discussion of bulk vs single cell RNA-seq

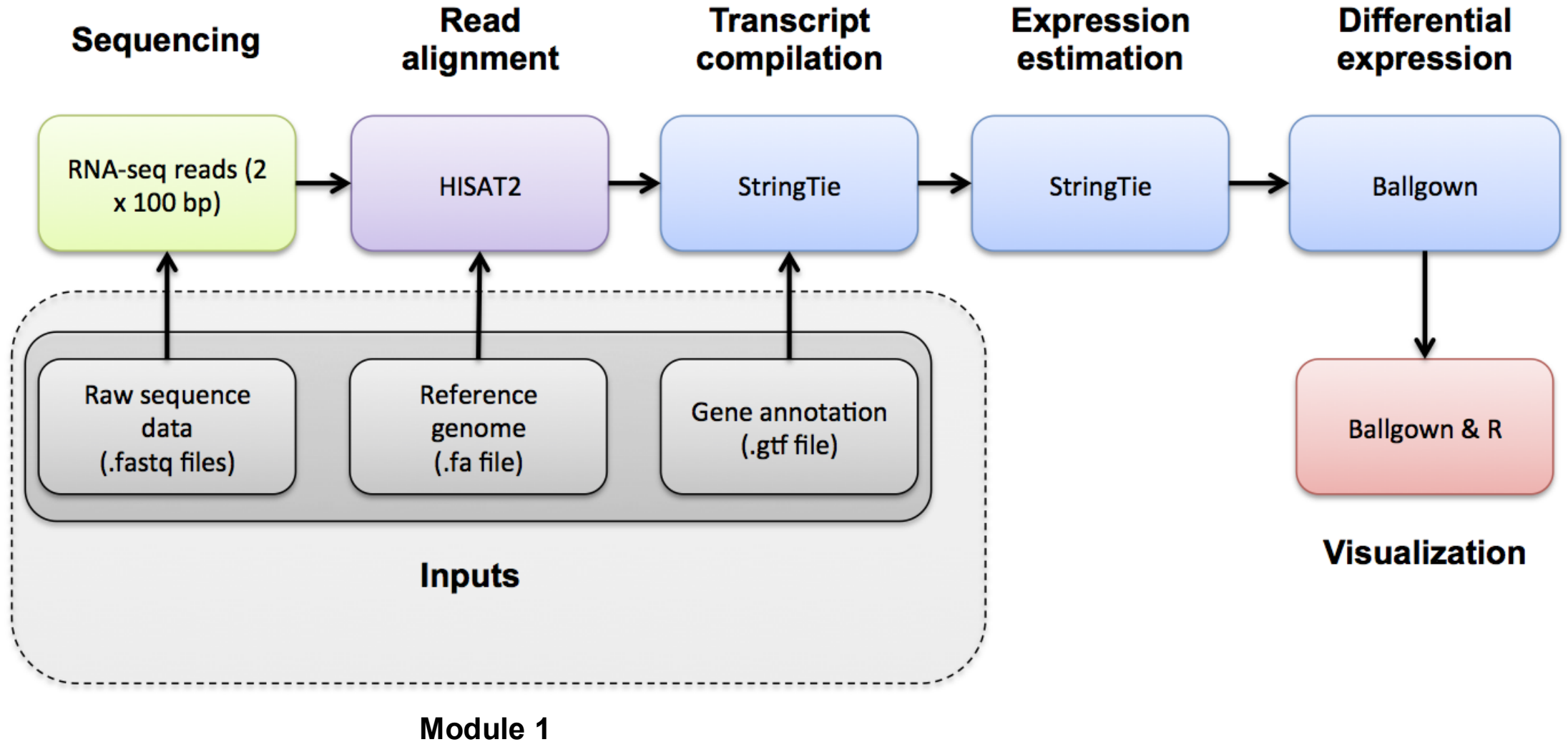


Factors to compare: Cost, complexity of library prep, complexity of analysis, qualitative and quantitative differences in richness of information obtained.



# **Introduction to tutorial (Module 1)**

# HISAT2/StringTie/Ballgown RNA-seq Pipeline



# Bioinformatics troubleshooting cheat sheet



- ❑ Check your inputs!
- ❑ Mix of incompatible reference genomes used (see [this tutorial](#))
- ❑ Mix of incompatible gene/transcript identifiers
- ❑ Reference sequence names (e.g. “1” vs “chr1”)
- ❑ 1-based vs 0-based coordinates (see [this tutorial](#))
- ❑ Computational tasks fail due to resource limitations (memory and storage)
- ❑ Dependency hell for bioinformatics tools. Learn to use containers (e.g. docker) or environment managers (e.g. conda)



# Coffee Break & Networking Session

## Workshop Sponsors

